

SYNOPSIS

At 1351 on 14 April 2007, the UK registered product tanker *Audacity* was involved in a collision with the Panama registered general cargo ship *Leonis*, in very poor visibility, in the precautionary area at the entrance to the River Humber. Both vessels sustained damage to their bows. Fortunately there were no injuries and no pollution was caused.

Audacity had been outward bound from Immingham Oil Terminal and was approaching the precautionary area in order to disembark her pilot. *Leonis* had entered the precautionary area from seaward and had just completed embarking her pilot.

The MAIB investigation found that the operation of the bridge team on *Audacity* was inadequate, and the extent of the VTS area and VTS powers was not clearly understood by the VTS operators.

The investigation identified contributing factors to the accident; these included:

- The pilots and bridge teams, on both vessels, did not make a full assessment of the risk of collision.
- VTS procedures for managing traffic in the precautionary area were insufficient.
- VTS operators were unaware of the poor visibility in parts of the VTS area.
- Humber VTS did not have a formal operating procedure for periods of reduced visibility.
- Communications were poor.
- The Port Authority misunderstood how risk assessment could be used to improve the effectiveness of the VTS operations.

As a result of this accident, Associated British Ports Humber Estuary Services (ABP HES) has taken several actions to improve the performance of the VTS, pilots and pilot boarding operations.

Recommendations have been made to: the UK Major Ports Group/British Ports Association regarding the situational awareness of pilots immediately prior to boarding vessels; and to the ABP Group to develop policies covering headline issues, which can be implemented and monitored in all ports within the Group.

1.2 BACKGROUND INFORMATION

1.2.1 *Audacity*

At the time of the accident, *Audacity* was owned by F.T Everard Shipping Ltd and operated by F.T Everard & Sons Ltd. A transfer of ownership and management to James Fisher and Sons Ltd was in progress.

Audacity had just spent 2 weeks in a dry dock at Sunderland, where scheduled maintenance and repairs had been undertaken. Work at this time included an upgrade to the vessel's Electronic Chart Display and Information System (ECDIS), however it had not been possible to commission the equipment and, as a consequence, the ECDIS was not in operation at the time of the accident.

On completion of the dry docking, *Audacity* was chartered to load a cargo of gas oil at Immingham for discharge at Aberdeen. The short voyage from Sunderland to Immingham was uneventful.

Audacity was fitted with a high lift rudder and was reported to be responsive to her helm down to speeds as low as 2.0 kts. She was also equipped with a controllable pitch propeller which would swing the bow to port when the pitch was placed astern.

Cargo operations at Immingham were completed early on the evening of 13 April, but sailing was delayed due to restricted visibility throughout the area.

1.2.2 *Leonis*

Leonis was owned by Prebble Properties Group Inc. and operated by MarPer S.A.M, based in Monaco, and traded worldwide. Prior to the accident, she had loaded bulk bauxite ore in Venezuela for discharge at Rotterdam and Grimsby. The passage from Venezuela had been uneventful and cargo operations at Rotterdam had gone smoothly.

Leonis left Rotterdam during the evening of 13 April, and arrived at the Humber outer anchorage the following morning.

1.2.3 Associated British Ports Humber Estuary Services - Vessel Traffic Services (Photograph 3)

Associated British Ports (ABP) was the Competent Harbour Authority for the Humber, as defined by the Pilotage Act 1987, at the time of the accident. However, many of the duties and responsibilities of ABP have been delegated to Associated British Ports, Humber Estuary Services (ABP HES) whose role was to provide vessel traffic services and an efficient pilotage service for the area.

VTS Humber operating service levels are designated in MSN 1796 (**Annex 1**). These levels are:

TOS – a traffic organisation service: A service to prevent the development of dangerous maritime traffic situations and provide for the safe and efficient movement of vessel traffic within the VTS area.

INS – an information service: A service to ensure that essential decision making information becomes available in time for onboard navigational decision making.

The requirement for a port to decide if there is a need to establish a VTS is contained in SOLAS Chapter V, regulation 12. Additional guidance in the operation of VTS is given in MGN 238, 239, 240 (**Annexes 2, 3 and 4**).

Photograph 3



VTS Humber

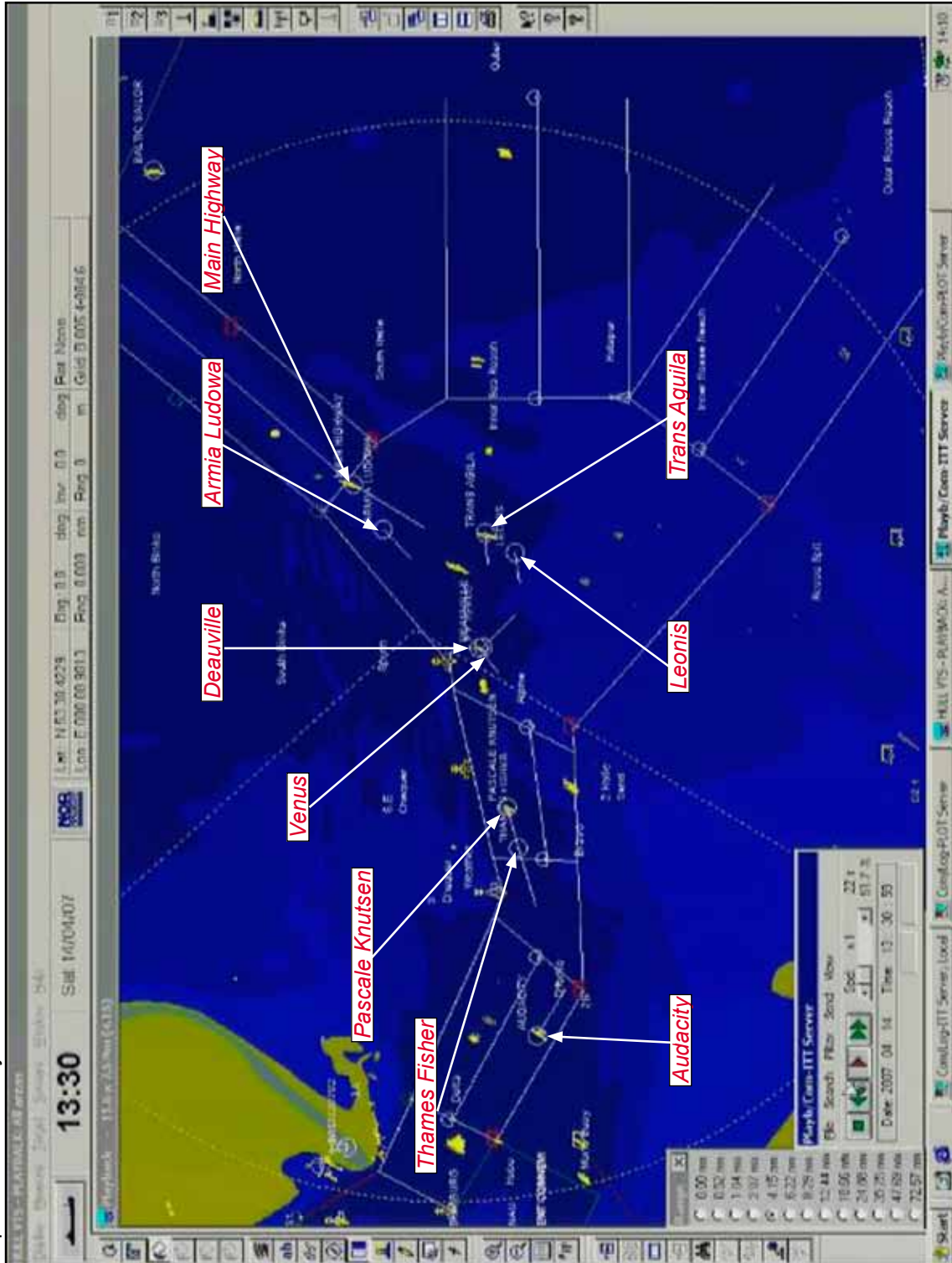
1.2.4 Traffic operating in the VTS area

There was a total of eight vessels operating in the area at the time of the collision, these were:

Vessel	Type	Designation	Pilot boarding time
<i>Thames Fisher</i>	Tanker	Inbound No1	1315
<i>Pascale Knutsen</i>	Tanker	Inbound No2	1315
<i>Deauville</i>	LPG	Inbound No3	1330
<i>Armia Ludowa</i>	Bulk	Inbound No4	1330
<i>Trans Aguila</i>	General	Inbound No5	1330
<i>Leonis</i>	Bulk	Inbound No6	1330
<i>Main Highway</i>	Car Carrier	Inbound Pilot Exempt	n/a
<i>Audacity</i>	Tanker	Outbound	already on board

The positions of these vessels at 1330 are shown on a chartlet of the VTS area (**Figure 1**). The turn allocated to each vessel to embark a pilot was determined both by time of arrival and the vessel's final destination. As can be seen at **Figure 1**, at 1330 *Armia Ludowa* and *Trans Aquila* are further from the pilot boarding area than *Leonis*, however they were scheduled to embark a pilot before *Leonis* due to the latter's shorter river transit.

Figure 1



VTS Humber screen shot at 13:30

Reproduced courtesy of ABP Humber

1.3 NARRATIVE

On the evening of 13 April, *Audacity* completed loading a full cargo of gas oil at Immingham Oil Terminal, berth No 8. Cargo calculations and documentation were completed and a pilot requested. On his arrival on board the vessel, the pilot informed the master that the visibility was too poor to allow *Audacity* to leave her berth.

The port regulations applicable to the Humber stated that, for vessels carrying dangerous cargo in bulk, a minimum visibility of 0.5 mile throughout the intended route was required before sailing from a berth within the port limits.

The pilot remained on the bridge of *Audacity* in the hope that there might be some improvement, but the visibility continued to worsen, until at 0200 the sailing was cancelled, awaiting improved visibility in the morning, and the pilot left the vessel.

During this period of poor visibility three restricted inbound vessels arrived in the VTS area. The VTS operator instructed the vessels to anchor until the visibility improved. Also at anchor, awaiting the tide, was the large bulk carrier *Armia Ludowa*.

At 0830 on 14 April, *Leonis* arrived within the port limits of the Humber from Rotterdam; at this time the visibility in the approaches to the anchorage was less than 3 cables. The master reported his arrival to the VTS operator and was advised to anchor his vessel and await the tide. He was also informed that a pilot would embark *Leonis* at 1330.

Visibility slowly improved during the morning until at 1100 VTS Humber resumed the movement of vessels carrying dangerous cargo. A pilot boarded *Audacity* at 1135 and there was time for a concise master / pilot exchange before unmooring commenced at 1148. The last mooring line was clear at 1152.

At 1204, after completion of unmooring and tidying up of the aft station, the second officer proceeded to the bridge to report for his watch. The master decided to send him back down to get his midday meal and told the second officer that he did not require him to return to the bridge until after the pilot had been disembarked. The result of this decision was that the bridge team of *Audacity* consisted of only the master and the pilot throughout the passage from the Immingham Oil Terminal to the precautionary area.

After clearing the berth, *Audacity* headed downriver and the master engaged auto pilot and ordered the engines full ahead. This gave a speed over the ground of about 10.0 kts. Master and pilot both used radar to monitor the vessel's progress, the master using the starboard radar display and the pilot the port. Both radars were the same make and model and equipped with ARPA facilities. However, neither the master nor the pilot used the radar's ARPA function to routinely acquire or track targets. Visibility at this time was about 1 mile.

Because of the prevailing poor visibility at the anchorage the master of *Leonis* decided to allow himself plenty of time to pick up the anchor and proceed at slow speed to the pilot boarding area 8.5 miles away. The anchor party began heaving the anchor at 1210, and by 1219 the anchor was away. At this time the master called Humber VTS to confirm the boarding time was still scheduled for 1330.

Once Humber VTS had decided the visibility in the area had improved sufficiently, the three vessels carrying dangerous cargo in bulk, which were waiting for pilots at the anchorage, were instructed to heave up their anchors in time to arrive in the pilot embarkation area between 1315 and 1330. The time that vessels would need to heave up anchor to arrive at the boarding area in accordance with this instruction was left to each master's discretion. At 1213, *Thames Fisher* reported to VTS Humber that they were underway, *Deauville* reported at 1220 and *Pascale Knutsen* at 1221.

Just after the last of these vessels reported they were underway, a report was made to the VTS operator, from a vessel anchored off Immingham, that the visibility in that area was less than 1 cable. The VTS operator acknowledged this report without further comment.

At 1228 *Trans Aguila* called VTS Humber to give 1 hour's notice of arrival at the pilot boarding area.

At 1233 *Thames Fisher* passed the "outer sea reach" reporting position and contacted the VTS operator, who cautioned that the vessel should not arrive at the pilot boarding area too early as her pilot was not scheduled to board until 1315.

At 1238 *Pascale Knutsen* reported in at the same position and the message was acknowledged without further comment. From available records it would appear that no report was made by *Deauville* when passing the "Outer Binks" reporting position or, *Leonis*, when she passed the "outer sea reach" reporting position.

At 1245 *Armia Ludowa* reported to VTS Humber that the vessel was underway from the anchorage, and a further report was made at 1306 when passing "Outer Binks".

At 1259 the VTS operator advised the pilot on board *Audacity* that pilot launch *Venus* had been assigned to conduct his disembarkation. However, the operator stated that as she had to embark pilots onto six vessels before that, she may be delayed. It was then left to the pilot's discretion to monitor VHF transmissions between the pilot launch and the vessels being boarded, and to adjust *Audacity's* speed to arrive after the last vessel boarding had been completed.

At 1303, a routine marine and safety information broadcast was transmitted on channel 12; this included navigation information and visibility reports, including that there was "nothing below 1 mile between Spurn Point and the Humber Bridge". Shortly after this broadcast ended, a vessel called the VTS on channel 12 to report there was reduced visibility in the Bull Channel, of between 100 to 200 metres.

At 1306, *Audacity* also reported "zero visibility" in the Bull Channel. In reply to *Audacity*, the VTS operator informed him that it was an isolated patch, and about 5 minutes later, visibility in the vicinity of *Audacity's* position improved to 6 or 7 cables.

In addition to the vessels that were required to embark and disembark pilots, the vehicle carrier *Main Highway* was proceeding inbound to the Humber and transiting the precautionary area. Her master held a pilotage exemption certificate (PEC) for the Humber and therefore was not required to embark a pilot before proceeding to her berth.

The sequence for placing pilots on the inbound vessels and taking the pilot off *Audacity* was set by the VTS operator. This relied on all six pilots boarding their respective vessels in sequence from the same pilot launch, *Venus*, during a 15 minute window, after which the launch would proceed to *Audacity* to disembark her pilot. In such situations, it was not routine for the VTS operator to inform vessels of the overall pilot boarding plan unless this was specifically requested. Accordingly, the master of *Leonis* was not specifically informed of the plan.

After entering the precautionary area at 1320, the master of *Leonis* informed the VTS operator that his vessel was 1 mile from the pilot boarding ground and asked for confirmation that a pilot would be available at 1330. In response, the VTS operator informed the master of *Leonis* that his pilot turn would be number six, that his vessel would embark a pilot after *Trans Aguila* and that he should reduce his speed “slightly”. The reason for this request was that *Trans Aguila* was astern of *Leonis*; however this was not explained to the master so he continued inbound keeping a safe distance ahead of the vessel on his starboard quarter, which was *Trans Aguila*.

At 1321 the first of the six inbound vessels, the tanker *Thames Fisher*, was approached by the pilot launch, and pilot embarkation operations commenced.

Despite being advised to reduce speed slightly, *Leonis* still arrived at the pilot station at precisely 1330, with *Armia Ludowa* and *Trans Aguila* yet to take their pilots (**Figure 1**). As there was no pilot available, the master decided to stop the engines and allow *Leonis* to run on to the west at about 5½ kts heading 282°, carried by the ship’s way and tidal stream.

Shortly after this time, *Audacity* was passing “Charlie” buoy, still proceeding at 10.0 kts, at which point she then began to reduce speed in preparation for her rendezvous with *P/L Venus*. Visibility had again reduced to about 1 cable, however there was no consideration given to summoning an additional officer or crew member to the bridge, nor was the sounding of fog signals commenced. No report on the poor visibility was made to VTS Humber.

At 1335, the master of *Leonis* realised his vessel was being quickly set towards the west south west, and he decided to turn *Leonis* to starboard, towards the north, in order to stem the tidal stream and maintain the vessel’s position. At this time, *Main Highway* was closing with *Leonis* as the vehicle carrier transited the precautionary area at about 8.2 kts. The master of *Leonis* initially misidentified the closing radar target as *Armia Ludowa*. Unsure about the intentions of the approaching vessel, the master of *Leonis* made a call to *Armia Ludowa* on VHF channel 13. *Armia Ludowa* responded, and it was agreed that the vessel would overtake to port of *Leonis*. Being satisfied with the arrangement, the master of *Leonis* allowed his vessel to continue its swing to starboard and then settle on a heading of 355°.

At 1338, the master of *Leonis* became concerned that the radar target of the vessel he had assumed to be *Armia Ludowa* indicated that the vessel was not taking the action as agreed during the earlier exchange by VHF. On comparing radar and AIS information, he realised that the approaching target was in fact *Main Highway*. He quickly contacted the master of *Main Highway* using VHF channel 13 who advised him

of his intention to pass to the north of *Leonis*. As a consequence, the master of *Leonis* placed the helm to port and, shortly afterwards, *Main Highway* passed to starboard at a distance of 2½ cables. This manoeuvre resulted in *Leonis* settling on a course of 245° with her engine stopped, which, combined with the effect of the tidal stream, caused her to move toward the exit of the outbound traffic lane.

By now, the fifth pilot had been embarked from the pilot launch, and *Venus* therefore began to head towards *Leonis*.

At 1341 the coxswain of *Venus* called the master of *Leonis* on VHF channel 13 and informed him that the pilot launch was approaching the vessel's port side from astern. The coxswain requested that the speed of *Leonis* be reduced to 6 kts to facilitate boarding of the pilot.

Before the pilot boarded *Leonis*, he consulted the pilot launch's radar. In accordance with the coxswain's preference, the radar's display was set on a 1-mile range with the ship's head up. In discussion with the coxswain, it was agreed that a target that was painting clearly at a range of about 0.9nm was the "Alpha" buoy. No outbound traffic was visible as *Audacity* was still more than 1 mile away at this stage.

At 1343 the coxswain of the pilot launch called *Leonis* by VHF to advise that the pilot was on board and that the vessel should "go back on course". This message had to be repeated because the master of *Leonis* was confused by the instruction. After receiving the repeated instruction to "go back on course" the master assumed the course to follow was his initial course, and brought *Leonis* round onto a heading of 282°, although she was still drifting under the tidal influence to the south west. Due to the confusion over the instruction to "go back on course" the master had missed the start of the message. He was therefore unaware that the pilot was on board and continued monitoring channel 13, awaiting further instructions.

When the embarkation of the pilot was complete, the coxswain reported this to the VTS operator, using Humber VTS private VHF channel – Ch 50. He then proceeded towards *Audacity* to disembark her pilot.

Audacity, proceeding at a reduced speed of 8.5 kts, passed the "Bravo" buoy at about the same time that the pilot boarded *Leonis*. The visibility in the area had reduced to 1-2 cables, and it was at this stage, with the vessels 2 miles apart, that the master on *Audacity* first began to systematically monitor the target of the approaching vessel. After the radar target of *Leonis* had been acquired, the master noted that an initial CPA of 2 cables to starboard was predicted. Therefore, the speed of *Audacity* was reduced to about 6 kts at the request of the pilot to provide more separation with the vessels which were embarking their pilots. But the effect of this manoeuvre on the CPA with *Leonis* was not checked by either master or the pilot.

As the pilot on board *Leonis* made his way to the bridge, *Audacity's* pilot attempted to call *Leonis* on VHF channel 14, but without reply. On reaching the bridge, *Leonis's* pilot became disoriented. The radars on the bridge were set up differently to his preferred settings, and he spent some time adjusting these before he realised that the vessel was positioned further south west in the channel than was normal and was much closer to No 2 Haille Sand buoy than he had expected from the orientation on board the pilot launch.

Once the pilot on *Leonis* had re-assessed the situation, his priority was to move her to the north, so he ordered the wheel to starboard and engines ahead - it was only then that he became aware of an outbound vessel.

He then reported to the VTS operator, advising him of his position and intention to head towards the north west to enter the inbound TSS. At the end of this report he asked what the outbound vessel's name was. The VTS operator informed him that it was *Audacity* and her pilot had been calling him on channel 14. Direct communication was established between the two pilots.

At the time the pilots began communicating with each other, the vessels were 5 cables apart and closing at a combined speed of about 10.0 kts. *Audacity's* master had realised that a close quarters situation was developing and that hand-steering was required. Instead of calling additional personnel to the bridge he engaged the hand-steering and placed himself on the wheel. This left the con solely with the pilot, without the support of a bridge team.

When the two pilots spoke to each other, *Audacity's* pilot informed *Leonis* that he was approaching the Haille Sand buoy and intended "to go up" (north) through the New Sand Hole. The pilot of *Leonis* then informed the pilot of *Audacity* that he thought he was "possibly right ahead" of him and that he was turning the vessel to starboard (north westerly) towards the "Alpha" buoy; there was a period of silence, after which *Audacity's* pilot said he would take his way off and "come down to the south".

Before he carried out the agreed actions, the pilot of *Audacity* became distracted by a radio exchange with the coxswain of the pilot launch about the arrangement for his disembarkation. The coxswain was informed that the pilot did not intend to leave *Audacity* until the vessel was clear of *Leonis*. Throughout this conversation no action was taken to increase the CPA with *Leonis*.

Very shortly after the conversation ended, *Audacity* and *Leonis* came in sight of each other and it was realised by both bridge teams that a collision was probable and imminent. Although the engines of both vessels were placed full astern, they collided, bow to bow, at 1351 in position Lat 53° 32.5 N Long 000° 13.6 W; this position is 2.5 cables NE of the No 2 Haille Sand port hand buoy, close to the exit from the eastbound lane of the traffic separation scheme. The combined speed at the time of impact was approximately 8.5 kts.

1.4 ENVIRONMENTAL CONDITIONS

Weather conditions at the time of the accident were as follows:

Wind	N x E 3
Sea / swell	Low and slight
Tide	1.5 – 2.0 kts from the north east
Visibility	Poor to very poor, fog patches

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

The hours of work and rest for those ships' officers, pilots and VTS operators involved in this accident were reviewed. However, fatigue is not considered a factor in this accident.

2.3 SIMILAR ACCIDENTS

From available records there have been six accidents within the VTS Humber area since 1998 which have relevance to this accident:

- 1998 – Grounding in VTS Humber area off Spurn Head
 - Relevance
 - MAIB made a recommendation to ABP Humber to encourage VTS operators to be more proactive and to provide timely warnings to vessels operating in an unsafe manner.
- 2004 – Grounding in VTS Humber area
 - Relevance
 - A vessel was manoeuvring to take a pilot when she grounded. No advice or warning was given by the VTS operator and the master found communications with the pilot launch confusing.
 - As a result of this accident, General Notice to Pilots 23/2004 was issued (**Annex 11**).
- 2005 – Collision in VTS Humber area in the Humber estuary
 - Relevance
 - This was a collision where the master, a PEC holder, was conducting navigation, alone, on the bridge.
 - As a result of the collision, the vessel's owner implemented a bridge procedure that required two navigation officers (one of which should be the master) to be present on the bridge at all times when in pilotage waters.
- 2005 – Collision between pilot boat and an anchored vessel
 - Relevance
 - After embarking a pilot on board a vessel at anchor in the Humber estuary the coxswain became disorientated in dense fog. After thinking himself to be well clear of the vessel he increased speed, but then struck the side of the vessel he had just left.

- 2006 – Hazardous incident (near miss) in precautionary area
 - Relevance
 - Two vessels were navigating within the precautionary area, one vessel had already dropped her pilot and was heading northwards when an inbound vessel coming from the TSS passed close.
 - VTS Humber was recommended by MAIB to carefully monitor incidents in the vicinity of the pilot boarding ground and, if necessary, re-assess its position.
- 2006 – Collision in the Humber Estuary
 - Relevance
 - Two vessels collided in the river below the Humber bridge, in dense fog.
 - As a result, HES issued Memo to Pilots (**Annex 12**) which gave advice to pilots in the conduct of their passage. It includes sections on safe speed, bridge teamwork, fog signals and the availability of the VTS operator assistance in case of pilot disorientation.
 - HES also undertook to review its risk assessment process.

2.4 VDR INFORMATION

In the case of *Audacity*, VDR is not a requirement under the regulations, and none was fitted. An ECDIS had been newly installed during the recent dry-docking, but had not been commissioned before the vessel departed.

The available VDR data from *Leonis* was incomplete due to incorrect commissioning of the equipment data inputs. This had resulted in no information being recorded from the helm, engine data or bridge microphones.

The managers of *Leonis* had developed extensive emergency procedures and checklists but none of these documents gave any guidance on the operation, maintenance or requirement for retaining VDR information, or under which circumstances VDR data should be saved.

Under SOLAS Chapter V – Safety of Navigation Regulation 20, it will become mandatory on 1 July 2008 for vessels over 20,000gt and constructed before 1 July 2002 to be fitted with a VDR or (S) VDR (**Annex 13** refers). On 1 July 2010 this requirement will be further extended to vessels from 3,000gt up to 20,000gt.

As such, a vessel's VDR, where fitted, should be considered "critical equipment" and therefore procedures for maintenance and use should be specified in the company SMS manuals. These should include instructions on the procedures for saving of VDR data, and its inclusion as a key item in emergency checklists.

2.5 HUMBER ESTUARY SERVICES

2.5.1 Port Marine Safety Code (PMSC) Audit

The port of Hull and Humber Estuary Services were part of the ABP group. During the investigation it was noted that the external compliance auditing of the Humber SMS was conducted in November 2005 by ABP personnel from within the group.

SECTION 3 - CONCLUSIONS

The following safety issues have been highlighted by the investigation.

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. The procedure for a pilot/coxswain briefing prior to embarking the vessel was not conducted efficiently. The radar equipment available in the launch was liable to severe shadow effect while close to vessels, making the identification of navigational markers unreliable. [2.11]

3.2 OTHER SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION ALSO LEADING TO RECOMMENDATIONS

1. From historical data, incidents in the Humber Estuary are occurring more frequently than weighted in their current risk matrix. This indicates the risk is greater than initially allowed for or that the safety barriers are insufficient or ineffective. [2.3 / 2.5.2]
2. There were no detailed marine policies applied throughout the group, which made the auditing of ports within the ABP group for compliance with the PMSC more difficult. [2.5.1]
3. Risk analysis should be reviewed as a matter of routine after any serious incident to ensure the effectiveness of the safety barriers or to evaluate the need for additional barriers. [2.5.1]

3.3 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE NOT RESULTED IN RECOMMENDATIONS BUT HAVE BEEN ADDRESSED

1. Due to a combination of circumstances the VTS operator allowed *Leonis* to drift into a dangerous position close to the exit from the outbound TSS. This action was compounded by the lack of traffic information to either *Leonis* or *Audacity* about the position of the other. [2.10.1 / 2.10.4]
2. *Main Highway's* transit of the precautionary area, at speed, and with substantial alterations of course during the pilot boarding operation, was not good seamanship, nor was it commented on by VTS. [2.8.1]
3. The powers of the AHM to give advice and guidance to vessels operating inside the VTS area, but outside the port limits, were not fully understood, and there was reluctance for operators to issue proactive information to vessels within the precautionary area. [2.6.1 / 2.10.1]
4. It was incumbent on VTS to ensure that its plan for boarding of pilots recognised the need for vessels arriving at the boarding area to be properly separated both geographically and in time. [2.6.2]
5. The VDR recording from *Leonis* was incomplete, and information regarding helm and engine status was not recorded. There were no procedures in the SMS for the use and maintenance of VDR equipment. [2.4]

6. Routine information broadcasts, including visibility reports, were made every 2 hours. Although several reports of reduced visibility were received, no formal re-assessment was made of the visibility in the estuary and no additional broadcasts were made. There were no formal reduced visibility procedures and no requirements for reduced visibility to be reported. [2.6.2]
7. Humber VTS had no formal procedures for the preservation of records in the event of an incident. [2.6.3]
8. *Leonis* altered course towards the north west because both master and pilot were unaware of the presence of *Audacity*. As a result, no assessment of the risk of collision was made before manoeuvring. [2.7.1 / 2.7.4]
9. ARPA was not used effectively on either vessel to assess risk of collision. By the time the ARPA was used on *Leonis*, it was too late for it to provide reliable information. [2.7.4 / 2.9.5]
10. Effectively, no-one held the con on the bridge of *Audacity* because both the master and pilot had deferred to the other, there was no discussion or questioning of the intentions of *Leonis*, and at a critical time they involved themselves with tasks that were inappropriate given the impending close quarters situation. [2.9.1 / 2.9.2]
11. The bridge on *Audacity* was insufficiently manned in the circumstances and conditions. It did not comply with company requirements or HES instructions to pilots, however no additional resources were requested by the pilot. [2.9.2]
12. Despite advising the pilot of *Leonis* that he would take action and come to the south, the pilot of *Audacity* did not alter course. This lack of action was not questioned by the master or the VTS operator, and the pilot of *Audacity* did not advise *Leonis's* pilot that he no longer intended to act as agreed. [2.9.2 / 2.10.3]
13. The communication between all parties involved was unclear and prone to misunderstanding, and use of standard marine phrases was not practised. [2.10]
14. VTS operators did not consider they were able to give advice and guidance to vessels with pilots on board. It was considered that the pilot would know what he was doing and that the operator did not need to be further involved once a pilot was on board. [2.10.2]
15. Communications from the VTS operator and *P/L Venus* were ambiguous and confusing. They were not result orientated and did not use identifier markers. Requests for specific information were inappropriately answered. [2.10.5 / 2.11]